# Brake Caliper Bracket Furnace Overview

| Field | Value |
|---|---|
| Furnace ID | HT-02 |
| Program | Brake Caliper Bracket |
| Standard Batch Size | 400 parts |
| Standard Batch Time | 240 min |
| Planned Batches/Week | 25 |
| Rated Weekly Output | 10,000 parts |
| Approved Alternate Furnace | None |
| Shared Capacity with Other Programs | 6% of planned time |

# Weekly Furnace Availability

| Week | Planned Hours | Unplanned Downtime Hours | Availability % |
|---|---|---|---|
| W14 | 120 | 14 | 88.3% |
| W15 | 120 | 18 | 85.0% |
| W16 | 120 | 21 | 82.5% |
| W17 | 120 | 16 | 86.7% |
| W18 | 120 | 19 | 84.2% |
| W19 | 120 | 15 | 87.5% |

# Unplanned Downtime Register

| Date | Event | Duration Hrs | Lost Batches Equivalent |
|---|---|---|---|
| W14 Tue | Thermocouple failure | 3.5 | 0.36 |
| W14 Thu | Quench pump reset | 2.0 | 0.21 |
| W15 Mon | Door seal trip | 1.5 | 0.16 |
| W15 Fri | Soak restart after interruption | 2.8 | 0.29 |
| W16 Tue | Conveyor indexing fault | 1.2 | 0.13 |
| W16 Wed | Thermocouple drift | 4.1 | 0.43 |
| W16 Sat | Quench agitation issue | 3.8 | 0.40 |
| W17 Thu | Door interlock reset | 2.6 | 0.27 |
| W17 Sat | Temperature overshoot check | 1.9 | 0.20 |
| W18 Tue | Quench pump reset | 2.4 | 0.25 |
| W18 Thu | Soak restart after trip | 3.6 | 0.38 |
| W18 Sat | Conveyor jam | 2.1 | 0.22 |
| W19 Wed | Thermocouple failure | 3.2 | 0.33 |
| W19 Fri | Door seal trip | 1.8 | 0.19 |

# Warm-Up and Stabilization Losses

| Week | Restart Events | Avg Warm-Up Loss per Restart Hrs | Weekly Warm-Up Loss Hrs |
|---|---|---|---|
| W14 | 3 | 1.2 | 3.6 |
| W15 | 4 | 1.2 | 4.8 |
| W16 | 4 | 1.3 | 5.2 |
| W17 | 3 | 1.2 | 3.6 |
| W18 | 4 | 1.2 | 4.8 |
| W19 | 3 | 1.1 | 3.3 |

# Cycle Interruptions and Reload Delays

| Week | Interrupted Batches | Avg Loss per Event Hrs | Reload Delay Hrs |
| --- | --- | --- | --- |
| W14 | 2 | 0.8 | 1.6 |
| W15 | 3 | 0.7 | 2.1 |
| W16 | 3 | 0.9 | 2.7 |
| W17 | 2 | 0.8 | 1.8 |
| W18 | 3 | 0.8 | 2.4 |
| W19 | 2 | 0.7 | 1.4 |

# Preventive Maintenance Exception

| Week | PM Window Planned | PM Window Taken | Deferred? | Reason |
|---|---|---|---|---|
| W14 | Yes | No | Yes | Volume recovery priority |
| W15 | No | No | No | NA |
| W16 | Yes | No | Yes | Dispatch recovery push |
| W17 | No | No | No | NA |
| W18 | Yes | No | Yes | Customer urgency |
| W19 | Yes | Partial | Yes | Weekend OT constraint |

# Shift Engineer Remarks

| Week | Remark |
| --- | --- |
| W14 | CNC stayed live while HT queue rose above floor limit. |
| W15 | Bracket lots held at furnace entry due to restart qualification. |
| W16 | Night shift lost 2.2 hrs after the trip and could not recover planned batches. |
| W17 | Machining made plans, dispatch still slipped due to HT carryover. |
| W18 | Repeated restart caused soak instability and late batch release. |
| W19 | Additional machining OT did not materially improve shipments. |

# Appendix: Other Furnace Family Assets

| Furnace ID | Program | Weekly Rated Capacity | Approved for Bracket? | Reason if No |
|---|---|---|---|---|
| HT-01 | Hub Carrier | 8,400 | No | Atmosphere mismatch |
| HT-03 | Clip Family | 12,000 | No | Fixture incompatibility |
| HT-04 | Tool Room | 1,600 | No | Lab furnace only |
| HT-05 | Pilot Cell | 2,000 | No | PPAP not approved |